Problem set 10

Question 2 Let $X_1, X_2, \ldots$ be random variables on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ Prove the following:

(a) $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X \iff \mathbb{E}\left[\frac{|X_n - X|}{1+|X_n - X|}\right] \xrightarrow{n\to\infty} 0$

(i) $\mathbb{E}\left[\frac{|X_n - X|}{1+|X_n - X|}\right] \xrightarrow{n\to\infty} 0$ if $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$.

$= \int$ let $Y_n = |X_n - X|$

$$\mathbb{E}\frac{|Y_n|}{1+|Y_n|}\, d\mathbb{P} = \int \frac{|Y_n|}{1+|Y_n|}\, d\mathbb{P}$$

$$= \int_{|Y_n|<\varepsilon} \frac{|Y_n|}{1+|Y_n|}\, d\mathbb{P} + \int_{|Y_n|\geq\varepsilon} \frac{|Y_n|}{1+|Y_n|}\, d\mathbb{P}.$$

So, $|Y_n| < \varepsilon \Rightarrow \dfrac{|Y_n|}{1+|Y_n|} \leq \dfrac{\varepsilon}{1+\varepsilon}$.

If $|Y_n| \geq \varepsilon \Rightarrow \dfrac{|Y_n|}{1+|Y_n|} < 1$

$$\therefore \mathbb{E}\frac{|Y_n|}{1+|Y_n|} \leq \frac{\varepsilon}{1+\varepsilon} \int_{|Y_n|<\varepsilon} d\mathbb{P} + \int_{|Y_n|\geq\varepsilon} d\mathbb{P}.$$

$$= \frac{\varepsilon}{1+\varepsilon} + \mathbb{P}(|Y_n| \geq \varepsilon).$$

given as $n \to \infty$ $\mathbb{P}(|Y_n| \geq \varepsilon) \to 0$. as $n \to \infty$.

since ε is arbitary,

$$\mathbb{E}\left[\frac{|Y_n|}{1+|Y_n|}\right) \longrightarrow 0.$$

(i) Given: $\mathbb{E}\left(\frac{|X_n - X|}{1+|X_n-X|}\right) \xrightarrow{n\to\infty} 0$ then

to show: $\mathbb{P}\left(\frac{|X_n-X|}{1+|X_n-X|} \geq \varepsilon\right) \Longrightarrow 0$ as $n \to \infty$.

if $\mathbb{E}\left(\frac{|X_n-X|}{1+|X_n-X|}\right) \xrightarrow{n\to\infty} 0$

$$\Rightarrow \frac{|X_n-X|}{1+|X_n-X|} = 0 \quad \text{a.s as } n\to\infty.$$

$$\Rightarrow |X_n - X| = 0 \quad \text{as. } n\to\infty.$$

$$\Rightarrow X_n \xrightarrow[n\to\infty]{a.s} X$$

$$\Rightarrow X_n \xrightarrow[n\to\infty]{o|P} X.$$

as a.s. convergence implies in probability convergence.

(ii) Given $X_n \xrightarrow[n\to\infty]{|P} X \Rightarrow \mathbb{E}\frac{|X_n-X|}{1+|X_n-X|} \longrightarrow 0.$

Let $Y_n = |X_n - X|$

$$\mathbb{E}\left[\frac{|X_n-X|}{1+|X_n-X|}\right] = \int_{\mathbb{R}} \frac{|X_n-X|}{1+|X_n-X|}\, d\mathbb{P}(\omega)$$

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By archimedian property, $\exists N \in \mathbb{N}$ s.t $\frac{1}{n} < \varepsilon$, $\forall n > N$

$$\Rightarrow \frac{1}{2^n} < \varepsilon.$$

$$\therefore \lim_{n\to\infty} \mathbb{P}\left(|X_n - X| > \frac{1}{2^n}\right) = 0.$$

$$\mathbb{E}\left[\frac{|X_n - X|}{1+|X_n-X|}\right] = \int_\Omega \frac{|X_n-X|}{1+|X_n-X|}\, d\mathbb{P}.$$

$$= \int \frac{|X_n-X|}{1+|X_n-X|}\, d\mathbb{P} + \int \frac{|X_n-X|}{1+|X_n-X|}\, d\mathbb{P}.$$

$\{\omega : |X-X_n| > \frac{1}{2^n}\}$ $\quad$ $\{\omega : |X_n-X| \le \frac{1}{2^n}\}$

$$\le \mathbb{P}\left(|X_n-X| > \frac{1}{2^n}\right) + \frac{\frac{1}{2^n}}{1+\frac{1}{2^n}}.$$

As $\mathbb{P}\left(|X_n-X| > \frac{1}{2^n}\right) \xrightarrow{n\to\infty} 0$

$$\lim \mathbb{E}\left[\frac{|X_n-X|}{1+|X_n-X|}\right] = \lim_{n\to\infty} \mathbb{P}\left(|X_n-X| > \frac{1}{2^n}\right) + \frac{\frac{1}{2^n}}{1+\frac{1}{2^n}}.$$

$$= 0 + \lim_{n\to\infty} \frac{1}{1+2^n}.$$

$$= 0.$$

$$\therefore \lim_{n\to\infty} \mathbb{E}\left[\frac{|X_n-X|}{1+|X_n-X|}\right] \xrightarrow{n\to\infty} 0.$$

(b) $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X \iff \mathbb{E}(|X_n-X| \wedge 1) \xrightarrow{n\to\infty} 0.$

(i) $\mathbb{E}(|X_n-X| \wedge 1) \xrightarrow{n\to\infty} 0$. then $\mathbb{E}(|X_n-X| \wedge 1) \xrightarrow[n\to\infty]{\mathbb{P}} 0.$

$\Rightarrow \min\{|X_n-X|, 1\} \xrightarrow{n\to\infty} 0$ a.s.

$\Rightarrow |X_n-X| \xrightarrow[a.s]{n\to\infty} 0$

$\Rightarrow |X_n-X| \xrightarrow[n\to\infty]{\mathbb{P}} 0.$

ii) if $\{|X_n-X| \wedge 1\} \xrightarrow[n\to\infty]{\mathbb{P}} 0$. then $\mathbb{E}(|X_n-X| \wedge 1) \xrightarrow{n\to\infty} 0$

By archimedian property, $\exists\, n \in \mathbb{N}$ s.t $\forall N \in \mathbb{N}$, s.t.

$\forall\, n \ge N$, $\frac{1}{2^n} < \frac{1}{n} < \varepsilon$.

$$\therefore \mathbb{P}\left(|X_n-X| > \frac{1}{2^n}\right) \xrightarrow{n\to\infty} 0.$$

$\mathbb{E}(|X_n - X|^1)$

$= \int_{\Omega} \{|X_n - X|^1\} \, d\mathbb{P}$.

$= \int_{\{\omega : |X_n - X|^1 \geq \frac{1}{2^n}\}} |X_n - X|^1 \, d\mathbb{P} + \int_{\{\omega : |X_n - X|^1 < \frac{1}{2^n}\}} d\mathbb{P}$.

$\leq \mathbb{P}(|X_n - X|^1 \geq \frac{1}{2^n}) + \frac{1}{2^n} \mathbb{P}(|X_n - X|^1 < \frac{1}{2^n})$.

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$\therefore \lim_{n \to \infty} \mathbb{E}(|X_n - X|^1)$

$\leq \lim_{n \to \infty} \mathbb{P}(|X_n - X|^1 \geq \frac{1}{2^n}) + \lim_{n \to \infty} \frac{1}{2^n} \mathbb{P}(|X_n - X|^1 < \frac{1}{2^n})$

$0 + \lim_{n \to \infty} \frac{1}{2^n} = 0$.

$\therefore \lim_{n \to \infty} \mathbb{E}(|X_n - X|^1) = 0$.

iii) (?)

$\inf\{\varepsilon > 0 : \mathbb{P}(|X_n - X| > \varepsilon) < \varepsilon\} \xrightarrow{n \to \infty} 0$.

$\Rightarrow X_n \xrightarrow[n \to \infty]{\mathbb{P}} X$

Sol: For r.v X_n & X, define:

$\eta_{X_n}(X_n, X) = \inf\{\varepsilon > 0 : \mathbb{P}(|X_n - X| > \varepsilon) < \varepsilon\}$

Question 3.

Proposition: $X_n \xrightarrow[n\to\infty]{a.s.} X$, iff $\forall \varepsilon > 0$,

$$\lim_{m\to\infty} \mathbb{P}\left(|X_n - X| \le \varepsilon, \forall n \ge m\right) = 1.$$

or equivalently, $\forall \varepsilon > 0$,

$$\lim_{m\to\infty} \mathbb{P}\left(\bigcap_{n=m}^{\infty} (|X_n - X| \le \varepsilon)\right) = 1.$$

(i) $X_n \xrightarrow[n\to\infty]{a.s.} X$.

i.e. $\exists N \subset \mathcal{F}$ ~~$\lim_{n\to\infty} X_n(\omega) = X$~~ . with $\mathbb{P}(N) = 0$, we have,

i.e. $\forall \omega \in N^c$, $\lim_{n\to\infty} X_n(\omega) = X$.

$\Rightarrow \forall \varepsilon > 0$, $\exists N(\varepsilon, \omega) \in \mathbb{N}$ s.t. $\forall n \ge N(\varepsilon, \omega)$

we have:

$$|X_n(\omega) - X| < \varepsilon.$$

$\forall n \ge N_\varepsilon$, $\mathbb{P}(|X_n(\omega) - X| < \varepsilon) = 1$.

$\Rightarrow \exists m \in \mathbb{N}$ s.t.,

$\forall n \ge m$, $\mathbb{P}(|X_n(\omega) - X| < \varepsilon = 1$.

$\Rightarrow \lim_{m\to\infty} \mathbb{P}\left(|X_n(\omega) - X| < \varepsilon, \forall n \ge m\right) = 1$.

$\therefore \lim_{m\to\infty} \mathbb{P}\left(|X_n(\omega) - X| < \varepsilon, \forall n \ge m\right) = 1$.

(ii) $\forall \varepsilon > 0$, $\lim_{m\to\infty} \mathbb{P}(|X_n - X| \le \varepsilon, \forall n \ge m) = 1$.

$$\Rightarrow \lim_{m\to\infty} \mathbb{P}\left(\bigcap_{n=m}^{\infty} \mathbb{P}(|X_n - X| \le \varepsilon)\right) = 1.$$

$$\Rightarrow \lim_{m\to\infty} \mathbb{P}\left(\bigcap_{n=m}^{\infty} \mathbb{P}(|X_n - X| \le \varepsilon)\right) \le \lim_{m\to\infty} \mathbb{P}\left(|X_n - X| < \varepsilon\right)$$

$\Rightarrow$ As $m \to \infty$, $n \to \infty$ ($\because n \ge m$).

$\Rightarrow \lim_{n\to\infty} \mathbb{P}(|X_n - X| < \varepsilon) = 1$.

$\lim_{n\to\infty} X_n(\omega) = X(\omega)$ a.s.

(b) $X_n \xrightarrow[n\to\infty]{a.s.} X$ iff $\forall \varepsilon > 0$,

$\lim_{m\to\infty} \mathbb{P}\left(|X_m - X_k| > \varepsilon, \text{ for some } n, k \geq m\right) = 0$.

(c)

From (a)

$X_n \xrightarrow[n\to\infty]{a.s.} X$ iff $\forall \varepsilon > 0$,

$\lim_{n\to\infty} \mathbb{P}(|X_n - x| \leq \varepsilon, \forall n \geq m) = 1$.

As a.s. convergence implies, in probability convergence,

$X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$ iff $\forall \varepsilon > 0$,

$\lim_{m\to\infty} \mathbb{P}(|X_n - x| \leq \varepsilon, \forall n \geq m) = 1$.

Taking compliment we have:

$\lim_{m\to\infty} \mathbb{P}(|X_n - x| > \varepsilon, \forall n \geq m) = 0$.

$\because |X_n - x| \leq |X_n - X_k| + |X_k - x|$

$\{|X_n - x| > \varepsilon\} \subseteq \{|X_n - x| > \frac{\varepsilon}{2}\} \cup \{|X_k - x| > \frac{\varepsilon}{2}\}$.

Now consider:

$$|X_n - X_k| \le |X_n - X| + |X - X_k|$$

Then:

$$\left\{|X_n - X_k| > \frac{\varepsilon}{2}\right\} \subseteq \left\{|X_n - X| > \frac{\varepsilon}{2}\right\} \cup \left\{|X_m - X_k| > \frac{\varepsilon}{2}\right\}.$$

$$\mathbb{P}\left(\sup_{n,k} |X_n - X_k| > \varepsilon\right) \le \mathbb{P}\left(|X_n - X| > \frac{\varepsilon}{2}\right) + \mathbb{P}\left(|X_m - X_k| > \frac{\varepsilon}{2}\right)$$

As ε is arbitary,

$$\lim_{k,n \to \infty} \mathbb{P}\left(|X_n - X_k| > \varepsilon\right) \le 0.$$

As $\mathbb{P}(\cdot) \ge 0$

$$\Rightarrow \lim_{n,k \to \infty} \mathbb{P}\left(|X_n - X_k| > \varepsilon\right) = 0. \qquad \text{QED.}$$

(a)(ii) $\forall \varepsilon > 0$,

$$\lim_{m \to \infty} \mathbb{P}\left(|X_n - X| \le \varepsilon, \forall n \ge m\right) = 1$$

$$\lim_{m \to \infty} \mathbb{P}\left(\bigcap_{n=m}^{\infty} |X_n - X| \le \varepsilon\right) = 1.$$

(b) $X_n \xrightarrow[n \to \infty]{a.s.} X$ iff $\forall \varepsilon > 0$,

$$\lim_{m \to \infty} \mathbb{P}\left(|X_n - X_k| > \varepsilon, \text{ for some } n, k \ge m\right) = 0.$$

From (a),

$$X_n \xrightarrow[n \to \infty]{a.s.} X \text{ iff } \forall \varepsilon > 0,$$

$$\lim_{m \to \infty} \mathbb{P}\left(|X_n - X| \le \varepsilon; \forall n \ge m\right) = 1 \quad \text{---} (1)$$

$$\therefore |X_n - X_k| < |X_n - X| + |X_k - X| \quad \text{---} (2)$$

Taking compliment of (2),

$$\lim_{n\to\infty} P(|X_n - X| > \varepsilon, \forall n \geq m) = 1.$$

$$\{|X_n - X_k| > \varepsilon\} \subseteq \{|X_n - X| > \tfrac{\varepsilon}{2}\} \cup \{|X_k - X| > \tfrac{\varepsilon}{2}\}.$$

$$P(|X_n - X_k| > \varepsilon, \forall n,k \geq m) \leq$$

$$P(|X_n - X| > \tfrac{\varepsilon}{2}, \forall n \geq m) +$$

$$P(|X_k - X| > \tfrac{\varepsilon}{2}, \forall k \geq m).$$

As ε is arbitary, we have:

$$\lim_{n\to\infty} P(|X_n - X| > \tfrac{\varepsilon}{2}, \forall n \geq m) = 0.$$

$$\lim_{k\to\infty} P(|X_k - X| > \tfrac{\varepsilon}{2}, \forall k \geq m) = 0.$$

$$\Rightarrow \lim_{m\to\infty} P(|X_n - X_k| > \varepsilon, n,k \geq m) \leq 0.$$

As $P(\cdot) \geq 0$.

$$\Rightarrow \lim_{m\to\infty} P(|X_n - X_k| > \varepsilon, n,k \geq m) = 0.$$